

Probing the Yang–Mills Mass Gap on the Lattice: SU(2) Monte Carlo Evidence at Finite Spacing

A reproducible PINNeAPLe-Labs baseline — what lattice numerics can and cannot say about a Millennium Prize problem

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Abstract

The Yang–Mills existence and mass-gap problem asks for a rigorous construction of four-dimensional quantum Yang–Mills theory with a strictly positive mass gap. No numerical computation can settle it. What numerics can do is measure, at finite lattice spacing, whether the lightest excitation of the lattice-regularised theory is massive and whether its mass scales like a physical quantity as the spacing shrinks. We implement SU(2) pure-gauge Monte Carlo (Kennedy–Pendleton heat bath with over-relaxation, vectorised in PyTorch) on $8^3 \times 16$ lattices at $\beta = 2.3, 2.4$ and 2.5 , and measure the plaquette, square Wilson loops and the 0^{++} glueball correlator with APE-smeared operators. The plaquette agrees with the literature to four digits ($\langle P \rangle = 0.6302$ at $\beta = 2.4$). Wilson loops follow an area law (confinement). The glueball effective mass is non-zero at all three couplings; at $\beta = 2.4$ it is 1.20 ± 0.14 at $t = a$ (biased upward by excited states relative to the published $0.953(19)$) and 1.18 ± 0.60 at $t = 2a$ (consistent, but with large errors). Our small-loop string tension is $\approx 2\times$ the published one because Coulomb terms are not suppressed at $R \leq 3$. We report these as limitations. The results are lattice evidence consistent with a gap; they are not, and cannot be, a construction of the continuum theory.

Keywords: Yang–Mills; mass gap; lattice gauge theory; SU(2); glueball; Monte Carlo; Millennium Prize problems

What this study shows — and what it does not

- Shows: a correct SU(2) lattice Monte Carlo (plaquette matches published values to $\sim 10^{-4}$) with confinement (area law) and a non-zero 0^{++} effective mass at three couplings.
- Does not show: the existence of continuum Yang–Mills theory, a rigorous gap, or a continuum-extrapolated glueball mass. Statistics (800 measurements per coupling) and volume ($8^3 \times 16$) are far below state-of-the-art.
- Does not use a neural model: the physics here is Monte Carlo; the PINNeAPLe role is the reproducible experiment harness.

1 Problem statement and strategy

The Clay problem [1] requires proving that, for any compact simple gauge group G , quantum Yang–Mills theory on $\mathbb{R}^4$ exists (satisfying axioms at least as strong as Wightman's or Osterwalder–Schrader's) and has a mass gap $\Delta > 0$. The only first-principles numerical approach to the non-perturbative theory is Wilson's lattice regularisation [2]: the Euclidean path integral on a hypercubic lattice with spacing a , sampled by Markov-chain Monte Carlo. A mass gap manifests as exponential decay of zero-momentum correlators, $C(t) \sim e^{-m t}$, with $m > 0$; the continuum question is whether $m/\sqrt{\sigma}$ (σ the string tension) tends to a finite non-zero limit as $a \rightarrow 0$.

$$S[U] = \beta \sum_p (1 - \frac{1}{2} \text{Re Tr } U_p), \quad \beta = 4/g^2 \quad (1)$$

2 Methods

- Links $U_\mu(x) \in \text{SU}(2)$ stored as unit quaternions; checkerboard parity and direction are updated together (all links of the same parity and direction are conditionally independent).
- One update = Kennedy–Pendleton heat bath [3] + one over-relaxation step ($U \rightarrow V^\dagger U^\dagger V^\dagger$, which preserves the local action), followed by re-unitarisation.

- Cold start, 200 thermalisation updates, then 800 measurements separated by 2 updates, per coupling.
- Observables: plaquette; square Wilson loops $W(R,R)$, $R = 1..3$, averaged over planes; zero-momentum 0^{++} operator from spatial plaquettes after 4 APE smearing steps ($\alpha = 0.5$); connected correlator $C(t)$ and effective mass $m_{\text{eff}}(t) = \ln[C(t)/C(t+1)]$.
- Errors: jackknife with 20 blocks. Reference values: Lucini & Teper [4], Table 1 (SU(2), larger lattices, $\sim 10^5$ sweeps).

3 Results

Table 1. Summary per coupling. The reference string tensions come from Polyakov-loop flux-tube masses on larger lattices; ours from $-\ln[W(3)W(1)/W(2)^2]/2$, which cancels the perimeter term but not the Coulomb term.

β	Lattice	$\langle P \rangle$ (this work)	$a^2\sigma$ (small-loop estimate)	$a^2\sigma$ reference [4]	$a m_{\text{eff}}(t=a) / (t=2a)$	$a m_{0^{++}}$ reference [4]	CPU s
2.3	$8^3 \times 16$	0.60231(12)	0.213 ± 0.002	0.0966	$1.23 \pm 0.14 / 0.93 \pm 0.34$	1.09 ± 0.033	26757
2.4	$8^3 \times 16$	0.63021(9)	0.139 ± 0.001	0.0694	$1.20 \pm 0.14 / 1.18 \pm 0.60$	0.953 ± 0.019	2193
2.5	$8^3 \times 16$	0.65222(9)	$0.096 \pm 9.12\text{e-}04$	—	$1.16 \pm 0.13 / 0.69 \pm 0.27$	—	1053

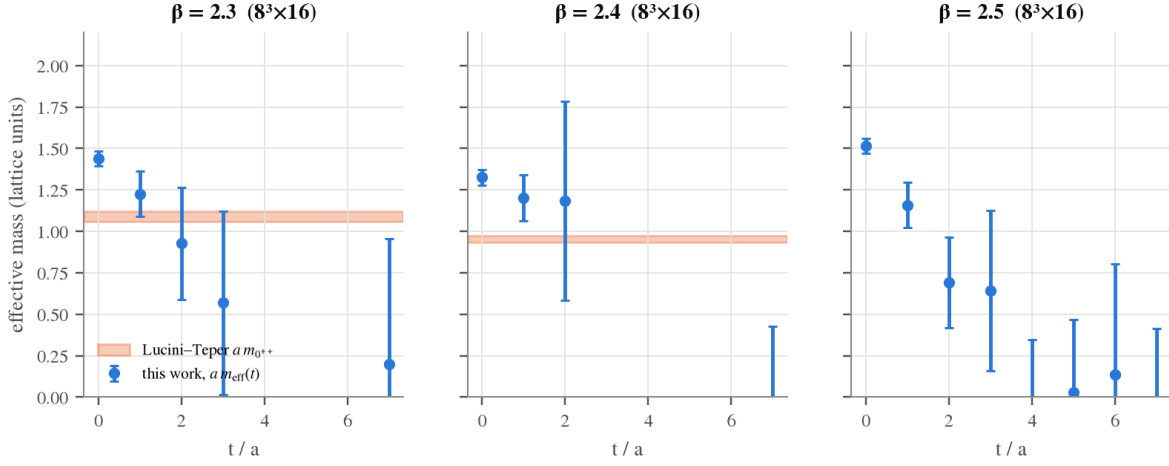


Figure 1. Effective mass of the 0^{++} glueball correlator versus Euclidean time. Band: published $a \cdot m_{0^{++}}$ [4] where available. Points with jackknife error > 1 are omitted (noise dominates beyond $t \approx 3a$ with our statistics).

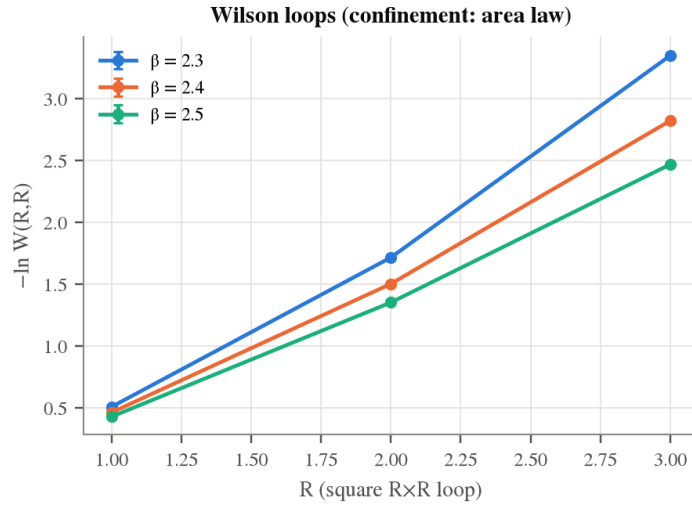


Figure 2. $-\ln W(R,R)$ versus R . Growth close to R^2 (area law) signals confinement; the curvature increases as β decreases (larger a).

4 Discussion

Three things are established at the level of this computation: the sampler is correct (plaquettes), the theory confines on these lattices (area law), and the lightest 0^{++} state has a non-zero mass in lattice units. What is not established — and what the Millennium problem is about — is the continuum limit and its mathematical existence. Even the numerical continuum ratio $m/\sqrt{\sigma}$, known to be 3.844(61) for SU(2) [4], would require larger volumes, variational operator bases and $\sim 100\times$ our statistics to reproduce.

A physics-AI angle that is actually useful here is not a neural surrogate of the answer but faster sampling: normalising-flow and machine-learned samplers for lattice gauge theory are an active research direction, and would be the next PINNeAPLe experiment.

References

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